COMPLEX HOTEL ENTERPRISE NETWORK ARCHITECTURE

EXECUTIVE SUMMARY

This document presents the complete technical architecture for the Complex Hotel Enterprise Network, designed and deployed over a six-week implementation period. The project delivers a fully redundant, segmented, and secure network infrastructure supporting 176 network devices across 140 guest rooms, 11 departmental VLANs, and multiple operational systems, including VoIP telephony, wireless connectivity, CCTV surveillance, and business-critical applications.

Key Project Achievements:

- Implementation of a three-tier hierarchical network design with full redundancy at core, distribution, and edge layers.
- Deployment of 11 VLANs providing logical segmentation for nine departments, guest services, and voice traffic.
- Establishment of dual-ISP connectivity with automatic failover via BGP and floating static routes.
- Installation of 140 access points with centralised wireless LAN controller management and seamless roaming capability.
- Integration of 140 IP phones utilising a centralised Call Manager Express (CME) deployment.
- Configuration of comprehensive access control lists (ACLs) to enforce inter-VLAN security policies.
- Achievement of 24/7/365 operational availability through redundant hardware, HSRP gateway failover, and OSPF dynamic routing.

Recommendations:

- Schedule bi-annual configuration backups and store offsite in an encrypted format.
- Implement network monitoring and logging solutions for proactive fault detection.
- Review ACL policies annually to align with evolving departmental access requirements.
- Plan for capacity expansion of guest VLANs to accommodate future occupancy growth.

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1. INTRODUCTION

1.1 Project Background

The Complex Hotel property required a complete overhaul of the network infrastructure to support modern hospitality operations, including high-density wireless connectivity for

guests, secure departmental segmentation, IP telephony services, and surveillance systems. The existing legacy network lacked redundancy, proper segmentation, and the capacity to support current and future technology demands.

1.2 Project Objectives

The primary objectives of this project were:

i. To Design and deploy a highly available, resilient enterprise network infrastructure supporting continuous 24/7/365 operation.
ii. To implement logical network segmentation for nine distinct departments, guest services, and voice communication.
iii. To provide seamless wireless coverage across all guest rooms, conference areas, and outdoor amenities, including pools and garden spaces.
iv. To integrate a centralised VoIP telephony system for internal and guest communication.
v. To establish dual Internet Service Provider connectivity with automatic failover capabilities.
vi. To enforce security policies through access control lists and device hardening measures.

1.3 Scope of Work

The project scope encompassed:

- Site survey, cabling infrastructure installation, and equipment racking.
- Configuration and deployment of core, distribution, and access layer switching infrastructure.
- Edge router configuration with BGP peering to two upstream ISPs.
- Wireless LAN controller setup and deployment of 140 access points.
- DHCP server implementation with redundant connectivity.
- IP phone registration and Call Manager Express configuration.
- Inter-VLAN routing, HSRP redundancy, OSPF dynamic routing, and NAT implementation.
- Comprehensive testing, documentation, and staff training.

2. DESIGN REQUIREMENTS

2.1 Business Requirements

Requirement Area	Operational Goal	Implementation Strategy
System Uptime	24/7/365 Service Continuity	Deploy redundant hardware and failover paths to eliminate single points of failure.
Guest Connectivity	Premium Wireless Experience	Ensure seamless roaming and high-density AP coverage across the entire property.
Process Stability	Uninterrupted Operations	Maintain a high-performance backbone for all critical business and staff functions.
Data Security	Traffic & Access Protection	Utilize advanced firewalls and encrypted tunnels to block unauthorized intrusion.
Voice Reliability	Internal & External Comms	Prioritize QoS (Quality of Service) for crystal-clear guest and staff telephony.
Elasticity	Future Growth Capacity	Build a modular infrastructure to scale with device count and bandwidth spikes.
System Convergence	Unified Tech Integration	Consolidate PMS, POS, and CCTV onto a single, managed network environment.

2.2 Technical Requirements

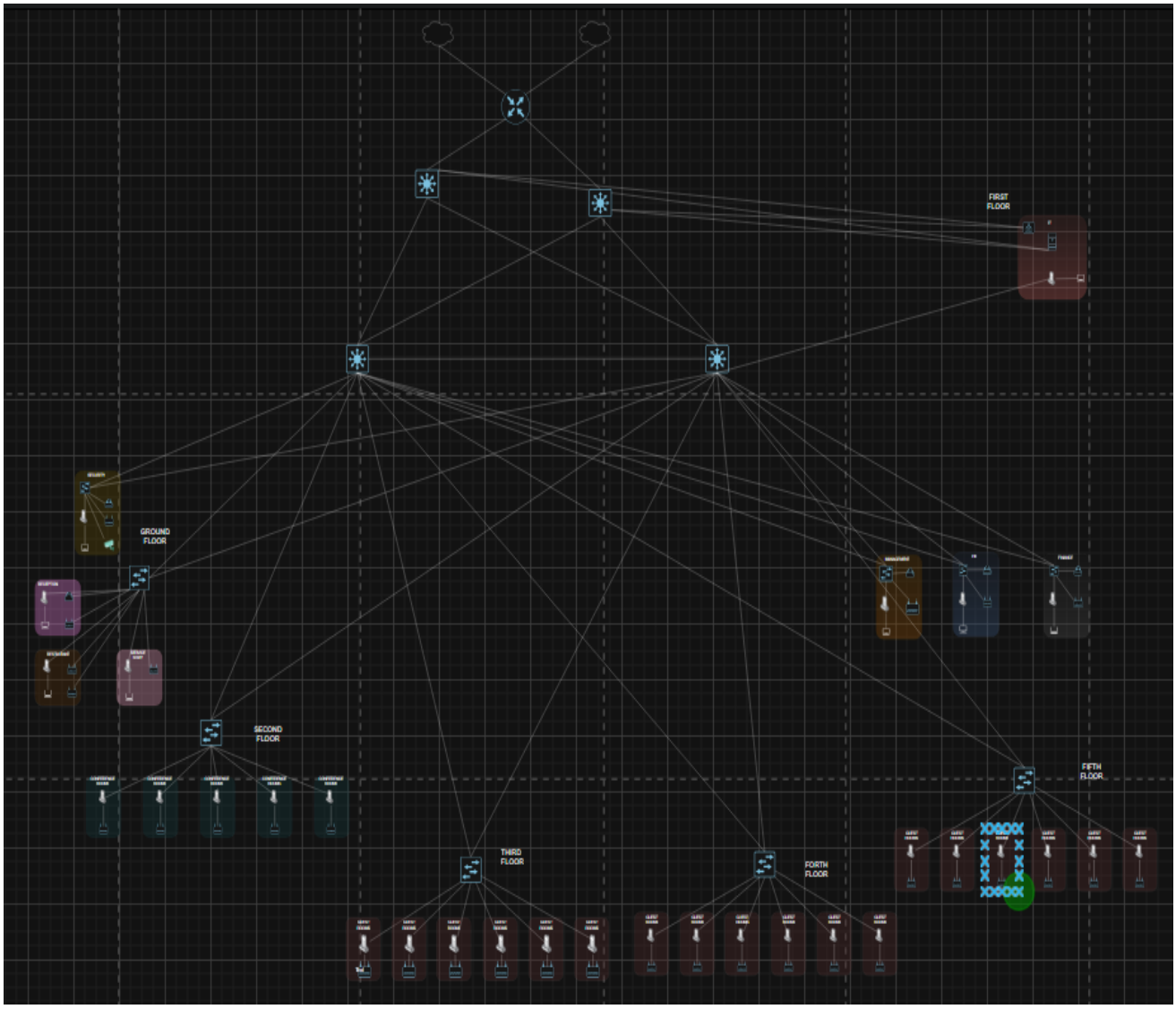
Infrastructure Component	Technical Implementation Details
ISP Redundancy	Dual ISP configuration using **BGP** and **floating static routes** for automatic failover.
Physical Cabling	Property-wide deployment of **CAT6 Ethernet** for in-room devices and APs.
Wireless Density	**140 in-room Access Points** (1 per guest room) to ensure maximum signal strength.
Mobility & Roaming	Wireless LAN Controller (WLC) with **FlexConnect** for seamless client transitions.
External Coverage	Strategic AP placement to provide full coverage for **pool and garden areas**.
Traffic Segmentation	**11 dedicated VLANs** established to isolate departments, CCTV, and guest traffic.
Device Power	**PoE-enabled** access switches deployed to power all peripheral network hardware.
Core Redundancy	**Dual-core and distribution switches** with **HSRP** for gateway redundancy.

3. NETWORK LAYOUTS (TOPOLOGY)

3.1 Physical Layout

The network implements a three-tier hierarchical model comprising Core, Distribution, and Access layers. Physical connectivity utilises CAT6 horizontal cabling for end-user devices and fibre backbone for inter-switch links where applicable.

Three-Tier Hierarchical Network Topology



Rack and Cabling Considerations:

i. Main Distribution Frame (MDF) houses Edge Router, L3SW1, L3SW2, WLC, and DHCP Server.
ii. Intermediate Distribution Frame 1 (IDF1) houses L3SW3 and access switches SW1, SW2, SW3, SW7, SW10.
iii. Intermediate Distribution Frame 2 (IDF2) houses L3SW4 and access switches SW4, SW5, SW6, SW8, SW9.
iv. All access switches connect via dual redundant 1 Gbps uplinks to both L3SW3 and L3SW4.

3.2 Logical Layout

Department	**VLAN ID**	**Network Address**	**CIDR**	**Gateway (HSRP VIP)**	**Subnet Mask**
Restaurant	10	192.168.0.0	/22	192.168.0.1	255.255.252.0
Conference Rooms	20	192.168.4.0	/22	192.168.4.1	255.255.252.0
Guest Rooms	30	192.168.8.0	/22	192.168.8.1	255.255.252.0
Security (CCTV)	40	192.168.12.0	/23	192.168.12.1	255.255.254.0
Service Staff	50	192.168.14.0	/23	192.168.14.1	255.255.254.0
IT	60	192.168.16.0	/24	192.168.16.1	255.255.255.0
Management	70	192.168.17.0	/24	192.168.17.1	255.255.255.0
HR	80	192.168.18.0	/24	192.168.18.1	255.255.255.0
Reception	90	192.168.19.0	/24	192.168.19.1	255.255.255.0
Finance	100	192.168.20.0	/24	192.168.20.1	255.255.255.0

Department	VLAN ID	Network Address	CIDR	Gateway (HSRP VIP)	Subnet Mask
Voice	200	172.16.0.0	/24	172.16.0.1	255.255.255.0

Routing and Redundancy:

i. OSPF Area 0 encompasses all routed links between the Edge Router, Core switches, and Distribution switches.
ii. HSRP provides first-hop redundancy for all user VLANs with L3SW3 as active (priority 110) and L3SW4 as standby (priority 100).
iii. BGP is configured on the Edge Router with eBGP peers to ISP1 (AS 10) and ISP2 (AS 20).
iv. NAT Overload (PAT) translates all internal traffic to the public IP addresses of the ISP connections.

4. INVENTORY LIST

4.1 Hardware Inventory

Hardware Component	Qty	Model Type	Functional Role
Edge Router	1	ISR (Integrated Services)	Internet Edge, BGP, NAT, and CME services.
Core Layer 3 Switch	2	Modular L3	High-speed backbone routing and core fabric.
Distribution Layer 3 Switch	2	Stackable L3	Inter-VLAN routing, HSRP, and ACL enforcement.

Hardware Component	Qty	Model Type	Functional Role
Access Layer Switch	10	PoE+ Access	End-user device connectivity and power delivery.
DHCP Server	1	Enterprise Server	Centralised IP address management and assignment.
Wireless LAN Controller	1	Enterprise WLC	Centralised AP management and radio frequency tuning.
Access Point	140	802.11ac Wave 2	Wireless client connectivity for guest rooms and facilities.
IP Phone	140	SIP PoE Phone	Property-wide voice communication via VoIP.
Network Printer	10	Laser Jet/MFP	Departmental document and administrative printing.
CCTV Camera	16	PoE IP Camera	Property surveillance and physical security monitoring.

4.2 Device Naming Convention

Device Category	Assigned Hostname(s)	Quantity	Administrative Role
Edge Router	EDGE	1	External Gateway & Perimeter Security
Core Switches	L3SW1, L3SW2	2	Network Backbone & High-Speed Switching

Device Category	Assigned Hostname(s)	Quantity	Administrative Role
Distribution Switches	L3SW3, L3SW4	2	Aggregation Layer & HSRP Redundancy
Access Switches	SW1 through SW10	10	End-Point Connectivity & PoE Delivery
DHCP Server	DHCP-SERVER	1	Dynamic IP Address Management
Wireless Controller	WLC	1	Centralised AP Management & Roaming

4.3 IP Address Allocations (Point-to-Point Links)

Interconnection Link	Network Address	CIDR	Purpose
Edge Router ↔ L3SW1	172.17.0.0	/30	Primary Edge-to-Core Transit
Edge Router ↔ L3SW2	172.18.0.0	/30	Secondary Edge-to-Core Transit
L3SW1 ↔ L3SW3	192.168.21.0	/30	Core-to-Distribution (Path A)
L3SW1 ↔ L3SW4	192.168.22.0	/30	Core-to-Distribution (Path B)
L3SW2 ↔ L3SW3	192.168.23.0	/30	Core-to-Distribution (Path C)
L3SW2 ↔ L3SW4	192.168.24.0	/30	Core-to-Distribution (Path D)
WLC ↔ Core Switches	192.168.25.0	/29	Wireless Management Subnet
L3SW1 ↔ DHCP Server	192.168.26.0	/24	Primary Server Access Path
L3SW2 ↔ DHCP Server	192.168.27.0	/24	Secondary Server Access Path

5. IMPLEMENTATION PLAN

5.1 Work Breakdown Structure

Week 1: Planning and Site Preparation

- Project kickoff and configuration review
- Site survey for AP placement and cable path planning
- Equipment receipt, inventory, and firmware consistency verification
- Rack layout design and MDF/IDF environmental checks

Week 2: Infrastructure Cabling

- Fibre backbone and horizontal cabling installation (Floors 1-4)
- Cable certification and patch panel termination
- Cable labelling and documentation

Week 3: Core and Distribution Deployment

- Rack and base configuration of L3SW1 and L3SW2
- OSPF configuration on core switches
- Rack and base configuration of L3SW3 and L3SW4
- SVI, HSRP, and OSPF configuration on distribution switches
- Edge Router configuration, including BGP and NAT
- Interconnect testing

Week 4: Access Layer Deployment

- Rack and configure SW1 through SW10
- PoE testing and VLAN assignment verification
- End-device connectivity testing

Week 5: Wireless and Voice Deployment

- WLC initialisation, interface, and WLAN (SSID) configuration
- Physical mounting of 140 access points
- AP registration and FlexConnect group configuration
- CME configuration on Edge Router
- IP phone registration (140 units) and voice quality testing
- Printer and CCTV connectivity verification

Week 6: Security, Testing, and Handover

- ACL verification and failover simulation testing (ISP, HSRP)
- Wireless roaming and load testing
- Security audit and vulnerability remediation
- Final documentation, configuration backup, and IT staff training
- Project sign-off

5.2 Week-by-Week Timeline

Week	Phase	Milestone Activities
Week 1	**Initiation & Prep**	Detailed site surveys, hardware inventory verification, and finalisation of the IP addressing schema.
Week 2	**Physical Layer**	Installation of CAT6 cabling, rack mounting equipment, and establishing physical connectivity.
Week 3	**Network Core**	Configuration of Core and Distribution switches (L3SW1–4), BGP on the Edge router, and HSRP setup.
Week 4	**Access Layer**	Deployment of SW1–SW10, VLAN trunking, and validation of PoE budget for APs and IP phones.

Week	Phase	Milestone Activities
Week 5	**Service Integration**	WLC provisioning, SSID broadcasting, and VoIP (CME) configuration for the 140 SIP phones.
Week 6	**Finalization**	Security hardening (ACLs), throughput testing, comprehensive documentation, and client handover.

6. SECURITY MEASURES

6.1 Access Control Lists (ACLs)

Comprehensive ACLs are applied inbound on each departmental VLAN SVI to enforce inter-VLAN traffic restrictions. The policy permits access to required shared services (WLC, DHCP) and the Internet while denying access to sensitive departmental VLANs.

ACL Summary:

VLAN	Department	ACL ID	Permitted Access	Restricted (Denied) Destinations
10	Restaurant	110	WLC, DHCP, Internet	Security, Staff, IT, Mgmt, HR, Reception, Finance
20	Conference	120	WLC, DHCP, Internet	Security, Staff, IT, Mgmt, HR, Reception, Finance
30	Guest	130	WLC, DHCP, Internet	**All internal VLANs** (Full Isolation)
40	Security	140	WLC, DHCP, Local Subnet	IT Department
50	Service Staff	150	WLC, DHCP, Internet	Security, Mgmt, HR, Finance, IT

VLAN	Department	ACL ID	Permitted Access	Restricted (Denied) Destinations
60	IT	160	**Any (Unrestricted)**	None (Full Management Access)
70	Management	170	WLC, DHCP, Internet	Staff, Reception, IT
80	HR	180	WLC, DHCP, Internet	Staff, Reception, IT
90	Reception	195	WLC, DHCP, Internet	Security, Mgmt, HR, Finance, IT
100	Finance	190	WLC, DHCP, Internet	Staff, Reception, IT

6.2 Device Hardening

All network devices implement baseline security hardening measures:

- Enable Secret Passwords: All devices configured with `enable secret class123`.
- Service Password Encryption: `service password-encryption` enabled globally.
- Login Banners: Warning banners configured on all console and VTY lines.
- Console Access Control: Console ports protected with local passwords.
- SSH Access: Edge Router configured for SSHv2 remote administration with local authentication.
- VTY Security: VTY lines restricted to SSH transport only on the Edge Router.

6.3 Network Segmentation

- Eleven distinct VLANs provide logical isolation between departments and guest networks.
- Voice VLAN (200) separates VoIP traffic from data traffic, enabling QoS prioritisation.
- Guest VLAN (30) is restricted from accessing any internal corporate VLANs via ACL 130.

- Sensitive departments (Finance, HR, Management) have restrictive ACLs allowing only essential services.

7. BACKUP AND RECOVERY

7.1 Redundancy and High Availability

The network design incorporates multiple layers of redundancy to ensure business continuity:

Infrastructure Component	**Redundancy Mechanism**	**Operational Benefit**
Internet Edge	Dual ISPs via **BGP** & **Floating Static Routes**	Automatic failover ensures zero downtime if a service provider goes offline.
Core Routing	Dual **L3SW1/L3SW2** with **OSPF** ECMP	Load balances traffic across the backbone and provides instant re-routing.
Default Gateway	**HSRP** between **L3SW3** and **L3SW4**	Clients maintain their gateway connection even if one distribution switch fails.
Distribution Layer	Dual switches with redundant Core uplinks	Prevents an aggregation layer failure from isolating entire sections of the hotel.
Access Connectivity	**Dual-homed uplinks** to the distribution layer	Ensures access switches (SW1-10) remain reachable if an uplink cable or port fails.
DHCP Services	Dual-homed server via separate Core subnets	Critical IP assignment services remain available across the routing fabric.
Wireless Control	WLC dual-connected via **/29 subnet**	Maintains centralised AP management and roaming through redundant core paths.

Infrastructure Component	Redundancy Mechanism	Operational Benefit
Power Resilience	**PoE** with **UPS-backed** building power	Keeps APs, Phones, and Cameras operational during short-term power fluctuations or outages.

7.2 Disaster Recovery Considerations

i. Configuration backups are performed at project completion and stored securely.
ii. In the event of a core switch failure, traffic automatically reroutes through the remaining core path.
iii. In the event of a distribution switch failure, HSRP ensures gateway failover within seconds.
iv. ISP failover occurs automatically via a floating static route with administrative distance 5 for the secondary ISP.

8. CONCLUSIONS AND RECOMMENDATIONS

8.1 Conclusions

The Complex Hotel Enterprise Network project has successfully delivered a fully redundant, secure, and scalable network infrastructure meeting all stated business and technical requirements. The three-tier hierarchical design with dual core and distribution switches, combined with OSPF dynamic routing and HSRP gateway redundancy, ensures the continuous availability of all hotel services. Logical VLAN segmentation, comprehensive ACL policies, and device hardening measures provide a robust security posture. Wireless coverage and VoIP services have been fully integrated and tested, providing a seamless guest and staff experience.

Key Success Metrics:

- All 176 network devices are deployed and operational.
- 140 access points providing full property wireless coverage.

- 140 IP phones registered and operational on centralised CME.
- All 11 VLANs are correctly segmented with ACL enforcement.
- Dual-ISP failover tested and validated.

8.2 Recommendations

i. Configuration Management: Implement automated configuration backup to a secure, off-site repository every quarter.
ii. Network Monitoring: Deploy SNMP-based network monitoring and syslog collection to proactively detect faults and performance degradation.
iii. ACL Review Cycle: Establish an annual review of ACL policies with department heads to ensure access permissions align with evolving operational needs.
iv. Capacity Planning: Monitor guest VLAN utilisation; the /22 subnet supports up to 1022 devices. Plan for VLAN expansion or subnet growth as occupancy or device density increases.
v. Wireless Survey Refresh: Conduct a post-deployment wireless site survey to validate coverage and identify any dead zones requiring additional AP placement.
vi. Firmware Lifecycle Management: Establish a scheduled maintenance window for firmware updates to address security vulnerabilities and feature enhancements.

VLAN & IP ADDRESSING SCHEME

Department	**VLAN**	**Network Address**	**Subnet Mask**	**CIDR**	**Gateway (HSRP VIP)**
Restaurant	10	192.168.0.0	255.255.252.0	/22	192.168.0.1
Conference Rooms	20	192.168.4.0	255.255.252.0	/22	192.168.4.1
Guest Rooms	30	192.168.8.0	255.255.252.0	/22	192.168.8.1
Security (CCTV)	40	192.168.12.0	255.255.254.0	/23	192.168.12.1

Department	VLAN	Network Address	Subnet Mask	CIDR	Gateway (HSRP VIP)
Service Staff	50	192.168.14.0	255.255.254.0	/23	192.168.14.1
IT	60	192.168.16.0	255.255.255.0	/24	192.168.16.1
Management	70	192.168.17.0	255.255.255.0	/24	192.168.17.1
HR	80	192.168.18.0	255.255.255.0	/24	192.168.18.1
Reception	90	192.168.19.0	255.255.255.0	/24	192.168.19.1
Finance	100	192.168.20.0	255.255.255.0	/24	192.168.20.1
Voice	200	172.16.0.0	255.255.255.0	/24	172.16.0.1

TESTING & VALIDATION PROCEDURES

Test ID	Testing Objective	Success Criteria (Expected Result)
TC-01	**Routing Fabric Integrity**	All OSPF neighbours reach **FULL** state across the Core and Distribution layers.
TC-02	**Gateway Failover (HSRP)**	Upon L3SW3 failure, **L3SW4** assumes the Active role within 10 seconds.
TC-03	**ISP Resiliency (BGP)**	Upon ISP1 failure, the default route shifts to **ISP2** within 15 seconds.
TC-04	**VLAN Segmentation**	ACLs correctly permit intended traffic and drop unauthorised inter-VLAN requests.
TC-05	**IP Management (DHCP)**	Clients in all VLANs (10-200) receive correct IP, Mask, and Gateway info.

Test ID	Testing Objective	Success Criteria (Expected Result)
TC-06	**Mobility & Roaming**	Wireless clients transition between APs with **zero packet loss** or session drops.
TC-07	**Voice Quality (QoS)**	VoIP calls maintain a **MOS score ≥ 4.0** with no jitter or audible artefacts.
TC-08	**Power Distribution**	All 296 PoE devices (APs, Phones, CCTV) show "On" and stable power draw.
TC-09	**Security Hardening**	Unauthorised ICMP/Ping requests across restricted VLAN boundaries fail as per policy.
TC-10	**Surveillance Integrity**	All **16 CCTV cameras** are reachable and provide stable video streams to the NVR.

BUDGET & COST ESTIMATES

Hardware Costs

Component	Qty	Unit Cost	Total Cost
Edge Router	1	$4,500	$4,500
Core & Distribution Layer 3 Switches	4	$6,000	$24,000
PoE+ Access Switches	9	$2,500	$22,500
Enterprise DHCP Server	1	$3,000	$3,000
Wireless LAN Controller (WLC)	1	$8,000	$8,000
802.11ac Wave 2 Access Points	140	$200	$28,000

Component	Qty	Unit Cost	Total Cost
SIP PoE IP Phones	140	$150	$21,000
Departmental Network Printers	10	$500	$5,000
PoE IP CCTV Cameras	16	$300	$4,800
CAT6 Infrastructure (Drops & Labor)	500	$50	$25,000
SUBTOTAL HARDWARE & CABLING			**$145,800**

Total Project Cost

Budget Category	Allocated Amount	Percentage of Total
Hardware Procurement	$145,800	44.8%
Professional Labor & Integration	$117,200	36.0%
Cabling & Infrastructure Materials	$25,000	7.7%
Racks, Cabinets, & PDUs	$8,000	2.4%
Project Contingency (10%)	$29,600	9.1%
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TOTAL ESTIMATED INVESTMENT	**$325,600**	**100%**

AS-BUILT DEVICE INVENTORY

Hostname	Device Type	Primary IP / Interface Addresses	VLAN Support	Physical Location
EDGE	Router	172.17.0.1, 172.18.0.1	Transit Only	MDF
L3SW1	Core Switch	172.17.0.2, 192.168.21.1, 192.168.22.1, 192.168.25.2, 192.168.26.1	Core Routing	MDF
L3SW2	Core Switch	172.18.0.2, 192.168.23.1, 192.168.24.1, 192.168.25.3, 192.168.27.1	Core Routing	MDF
L3SW3	Distribution	Multiple SVI Gateways (HSRP Active/Standby)	10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 200	IDF1
L3SW4	Distribution	Multiple SVI Gateways (HSRP Active/Standby)	10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 200	IDF2
SW1	Access	*Management IP TBD*	40 (Security), 200 (Voice)	IDF1
SW2	Access	*Management IP TBD*	10, 50, 90, 200 (Ops/Staff/Rec)	IDF1
SW3	Access	*Management IP TBD*	20 (Conf), 200 (Voice)	IDF1
SW4–6	Access	*Management IP TBD*	30 (Guest), 200 (Voice)	IDF2
SW7	Access	*Management IP TBD*	80 (HR), 200 (Voice)	IDF1
SW8	Access	*Management IP TBD*	70 (Mgmt), 200 (Voice)	IDF2

Hostname	Device Type	Primary IP / Interface Addresses	VLAN Support	Physical Location
SW9	Access	*Management IP TBD*	100 (Finance), 200 (Voice)	IDF2
SW10	Access	*Management IP TBD*	60 (IT), 200 (Voice)	IDF1
WLC	Controller	192.168.25.4, 192.168.25.5	Wireless Mgmt	MDF
DHCP	Server	192.168.26.2, 192.168.27.2	IP Services	MDF